Department of Electronics and Electrical Engineering
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EE101 - Basic Electronics

Tutorial 6: Solutions **Date: Sep 14, 2023**

1. Select the values of the resistors R_1 and R_2 in the circuit shown in Figure 1 such that $v_R(0^+) = 10V$ and $v_R(1\ ms) = 5V$.

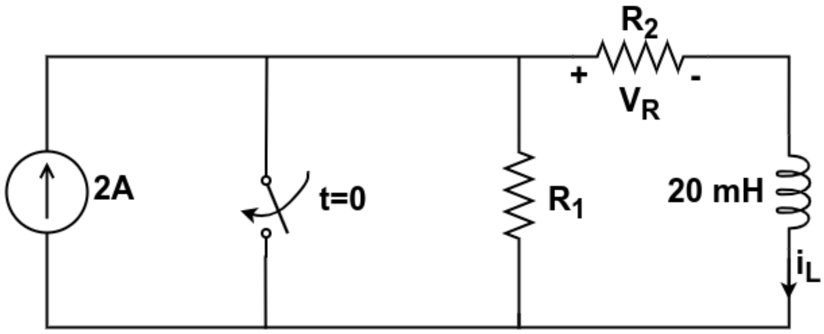


Figure 1 Electric circuit with an independent current source, a switch, resistors and an inductor. The switch closes at $t = 0$.

Solution

The switch is closed at $t = 0$. At $t = 0^+$, the voltage across R_2 is $v_R(0^+) = 10V$. Thus, the inductor current is $i_L(0^+) = \frac{v_R(0^+)}{R_2} = \frac{10}{R_2}$. As the switch is closed at $t = 0$, the decaying inductor current $i_L(t)$ bypasses R_1. Hence, $i_L(t)$ passes through R_2 only. It can be observed that $i_L(\infty) = 0$. Thus, $i_L(t)$ can be written as follows.

$$i_L(t) = i_L(0^+)exp\left(-\frac{t}{\tau_{LR}}\right) \quad (1)$$

Here, the time constant τ_{RL} for the series RL circuit can be computed as $\tau_{RL} = \frac{L}{R_2} = \frac{20\times10^{-3}}{R_2}$.
The voltage across R_2 for $t > 0$ can be written as follows.

$$v_R(t) = R_2 \times i_L(0^+)exp\left(-\frac{t}{\tau_{LR}}\right) \quad (2)$$

Thus, at $t = 1ms$ we have the following.

$$\begin{aligned} v_R(1\times10^{-3}) &= = R_2 \times \frac{10}{R_2}exp\left(-\frac{R_2\times1\times10^{-3}}{20\times10^{-3}}\right) \\ 5 &= 10exp\left(-\frac{R_2}{20}\right) \quad (v_R(1ms) = 5V) \\ R_2 &= -20ln(0.5) \end{aligned}$$

The above equations provide us with $R_2 = 13.8629\Omega$. Thus, $i_L(0^+) = \frac{V_R(0^+)}{R_2} = \frac{10}{13.8629} = 0.7213A$. Thus, $i_L(0^-) = i_L(0^+) = 0.7213A$. At $t = 0^-$ (before switch operation), the current through R_1 is $(2 - 0.7213)A$ or $1.2787A$. The inductor is considered as a short at $t = 0^-$. Hence, the voltage across R_1 is also $10V$. Thus, $1.2787 \times R_1 = 10$ or $R_1 = 7.8204\Omega$.

Hence, we have $R_1 = 7.8204\Omega$ and $R_2 = 13.8629\Omega$

2. Construct a negative edge triggered JK Flip-flop using a positive edge triggered T Flip-flop and logic gates.

Solution

The required inputs to the T Flip-Flop corresponding to the inputs of the JK Flip-Flop can be seen as follows:

Q_n	J	K	T	Q_{n+1}
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

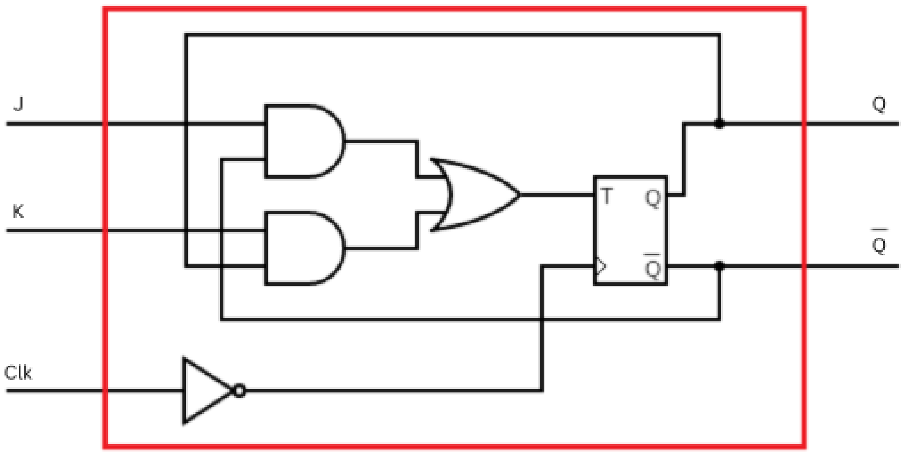
The Karnaugh Map for T as a function of Q_n, J and K is as follows:

Q_n \ JK	00	01	11	10
0	0	0	1	1
1	0	1	1	0

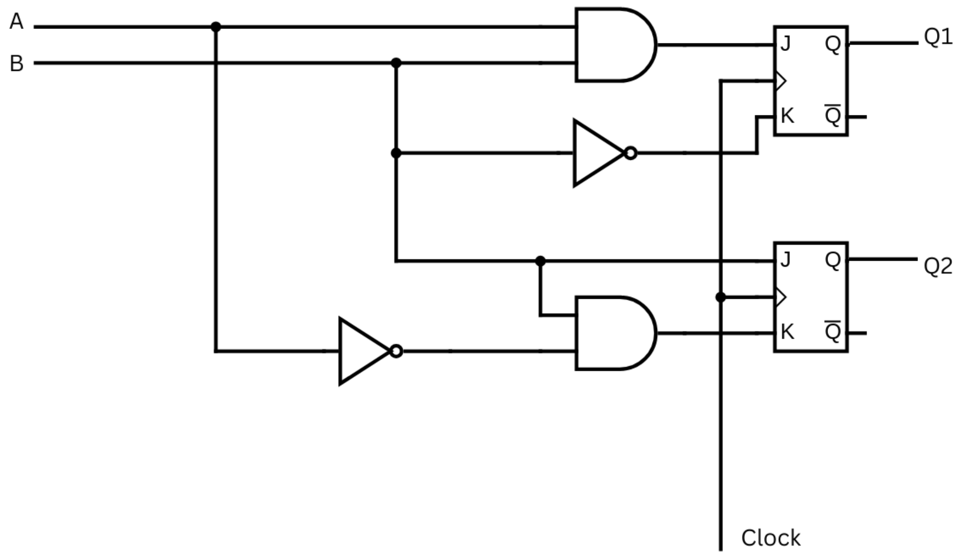
The minimal SOP form is

$$T(Q_n, J, K) = J.\overline{Q}_n + K.Q_n$$

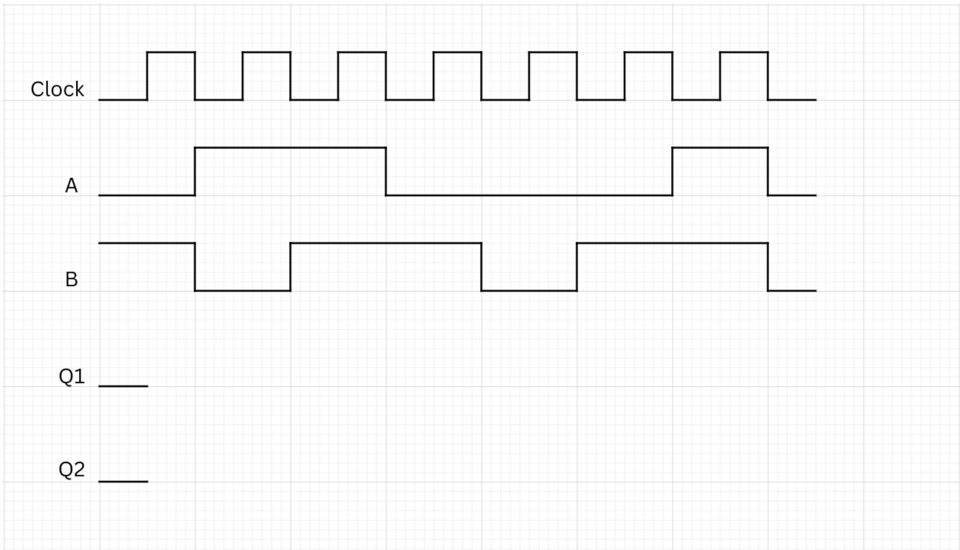
This can be implemented using two AND gates and an OR gate. The T Flip-flop can be triggered by the negative edge of the clock using a NOT gate. The circuit diagram for this is shown below:



3. Consider the circuit shown in the figure below.

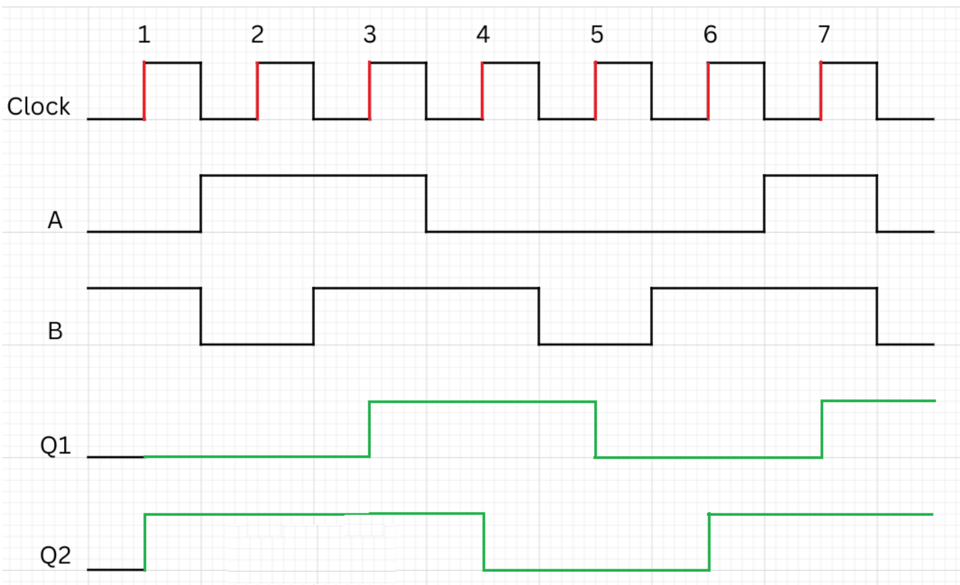


Plot $Q1$ and $Q2$ for the inputs shown below.



Solution

Plots of $Q1$ and $Q2$ can be obtained as shown below.

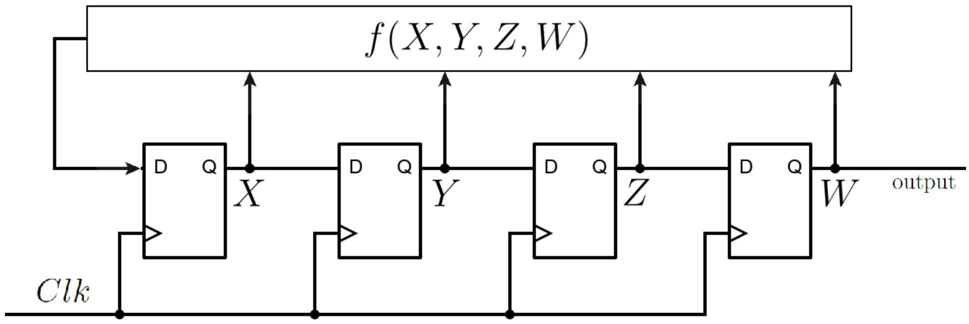


In the figure above, the triggers for the positive edge triggered FFs are shown in red. Plots of $Q1$ and $Q2$ are in green. The inference regarding the evolution of the values of $Q1$ and $Q2$ can be performed as shown in the table below.

Note that $J1 = A.B$, $K1 = \overline{B}$, $J2 = B$ and $K2 = \overline{A}.B$.

Trigger No.	Inputs		Flip-Flop 1				Flip-Flop 2			
	A	B	$Q1$	$J1$	$K1$	$Q1$	$Q2$	$J2$	$K2$	$Q2$
1	0	1	0	0	0	0	0	1	1	1
2	1	0	0	0	1	0	1	0	0	1
3	1	1	0	1	0	1	1	1	0	1
4	0	1	1	0	0	1	1	1	1	0
5	0	0	1	0	1	0	0	0	0	0
6	0	1	0	0	0	0	0	1	1	1
7	1	1	0	1	0	1	1	1	0	1

4. Consider the sequential circuit shown in the figure below, consisting of 4 D Flip-flops. Let the contents of the Flip-flops be denoted by X, Y, Z and W. Given that the function $f(X,Y,Z,W) = Z \oplus W$, determine the State Equations and the State Table for the circuit. Assuming that the initial state is $XYZW = 1100$, determine the output sequence.



Solution

Consider the circuit before the $n+1$-st clock strike (+ve edge of the clock signal). The state of the Flip-Flops can be denoted as X_n, Y_n, Z_n and W_n. The input to the D Flip-Flops are $Z_n + W_n$, X_n, Y_n and Z_n respectively (from left to right).

The State Equations can be written as:

$$\begin{aligned} X_{n+1} &= Z_n + W_n \\ Y_{n+1} &= X_n \\ Z_{n+1} &= Y_n \\ W_{n+1} &= Z_n \end{aligned}$$

The State Table can be obtained as:

Present State				Next State				Output
X_n	Y_n	Z_n	W_n	X_{n+1}	Y_{n+1}	Z_{n+1}	W_{n+1}	W_n
0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0
0	0	1	0	1	0	0	1	0
0	0	1	1	0	0	0	1	0
0	1	0	0	0	0	1	0	0
0	1	0	1	1	0	1	0	0
0	1	1	0	1	0	1	1	0
0	1	1	1	0	0	1	1	0
1	0	0	0	0	1	0	0	0
1	0	0	1	1	1	0	0	0
1	0	1	0	1	1	0	1	0
1	0	1	1	0	1	0	1	0
1	1	0	0	0	1	1	0	0
1	1	0	1	1	1	1	0	0
1	1	1	0	1	1	1	1	0
1	1	1	1	0	1	1	1	0

If the starting state is 1100, the sequence of states that follow are 0110, 1011, 0101, 1010, 1101, 1110, 1111, 0111, 0011, 0001, 1000, 0100, 0010, 1001, 1100, 0110, 1011, 0101, ...

The corresponding output sequence is 0011010111100010011....

The output has a pattern of length 15 that repeats. This output is an example of a pseudo noise sequence.